Practice workbook · Interpret a raised or lowered blood cell count

Blood test results or ratios where white cell or platelet numbers differ from normal.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-ABNORMAL-BLOOD-CELL-COUNTS

1. Question 5 · part a.ii · 2 marks

November 2024 Paper 1

Worked-example retry · Attempt part a.ii only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

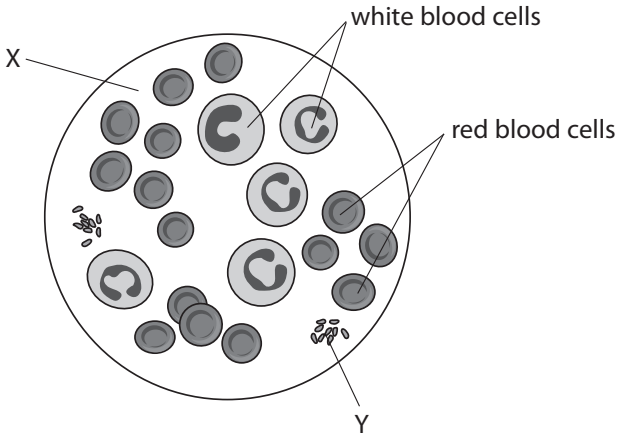
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2. Question 1 · part c.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.ii only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

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(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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2. Question 1 · part c.ii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.ii only.

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(ii) Suggest why this ratio will change if a person has a bacterial infection.

(2)

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(Total for Question 1 = 7 marks)

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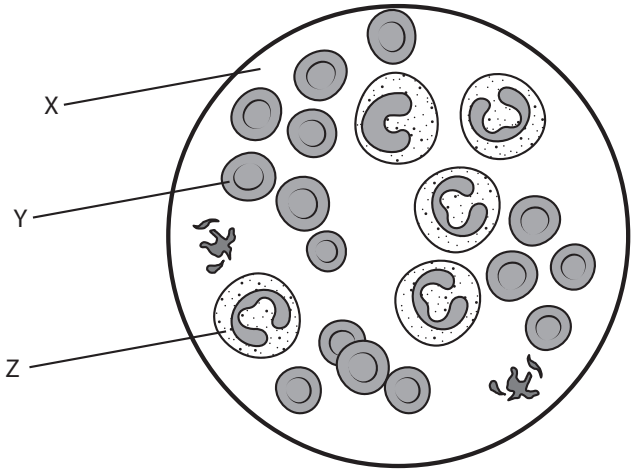
Working and answer

3. Question 2 · part a.iii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

(2)

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(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

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5

Turn over ▶

3. Question 2 · part a.iii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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4. Question 5 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

5 (a) The table gives information about some blood components in three people.

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(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
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1 : .....

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(2)

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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4. Question 5 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

(b) (i) Haemolytic anaemia (HA) is an inherited condition.
People with this condition have a low number of red blood cells in their blood.

Explain one symptom that people with HA will have.

(4)

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4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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